A Mini-Project Report on

Snake Game Implementation Using FPGA

Submitted by

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in partial fulfillment of the requirement for

TE (Semester-VI), Mini Project-2B

in

Department of Electronics & Telecommunication Engineering

under guidance of

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St. Francis Institute of Technology, Mumbai
An Autonomous Institute, Affiliated University of Mumbai

2025–2026

Certificate

This is to certify that **Bhavya Jadav, Abhinav Joshi, Aabhas Kasare,** and **Harshit Kawa** are the bonafide students of St. Francis Institute of Technology, Mumbai. They have successfully carried out the Mini Project-2B titled **"Snake Game Implementation Using FPGA"** in partial fulfillment of the requirement for **TE (Semester-VI)**, in Electronics and Telecommunication Engineering of Mumbai University during the academic year 2025–2026.

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Acknowledgement

We are thankful to a number of individuals who have contributed towards our third-year project. Without their help, this endeavor would not have been possible.

Firstly, we offer our sincere thanks to our project guide,**Dr.Ravindra Chaudhari**, for his constant and timely help and guidance throughout our preparation. We are also grateful to **Dr. Kevin Noronha**, Head of the Department (EXTC), for providing us with the necessary facilities and a conducive environment to work in.

Special thanks to the lab assistants and staff of the Electronics and Telecommunication Department for their technical support with the FPGA hardware. Finally, we would like to thank our parents and friends for their continuous encouragement and support during the completion of this project.

(Bhavya Jadav) (Abhinav Joshi)

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Abstract

This project presents the design and implementation of the classic Snake game on a Field-Programmable Gate Array (FPGA) using Verilog Hardware Description Language. The system is deployed on the Basys 3 FPGA board, which features the Xilinx Artix-7 chip, and outputs graphics to a VGA monitor at 640×480 resolution. The game allows the player to control a snake that grows longer as it consumes food items, with the objective of achieving the highest score possible without colliding with the walls or the snake's own body.

The hardware core is designed entirely in Verilog and comprises several interconnected modules: a clock divider for generating the 25 MHz VGA pixel clock from the board's 100 MHz oscillator, a VGA controller for generating synchronization signals and pixel coordinates, a game logic module for managing snake movement and state, a collision detection module, a pseudo-random food generator, and a score display module using the on-board seven-segment display. The direction of the snake is controlled using the four push buttons available on the Basys 3 board.

The project successfully demonstrates the use of FPGA-based digital design for real-time interactive applications. It provides hands-on experience in HDL programming, state machine design, VGA interfacing, and hardware-software co-design methodology using the Xilinx Vivado toolchain.

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List of Abbreviations

Abbreviation	**Full Form**
FPGA	Field-Programmable Gate Array
HDL	Hardware Description Language
VGA	Video Graphics Array
VHDL	VHSIC Hardware Description Language
VHSIC	Very High Speed Integrated Circuit
RGB	Red Green Blue Colour Model
FSM	Finite State Machine
CLK	Clock
HSYNC	Horizontal Synchronisation Signal
VSYNC	Vertical Synchronisation Signal

Chapter 1
Introduction

The Snake game is one of the most iconic arcade games in computing history. In this project, we implement a fully functional Snake game on a Basys 3 FPGA board using Verilog HDL, with output rendered on a VGA monitor. The game demonstrates a wide range of digital design concepts, including clock management, synchronous finite state machines, VGA signal generation, and modular Verilog design.

FPGAs offer a unique platform for such projects: unlike microcontrollers or software-based solutions, FPGAs allow true hardware parallelism. Every game module — collision detection, VGA rendering, food generation, and score tracking — executes concurrently in dedicated hardware logic, resulting in a highly deterministic, real-time system with no operating system overhead.

1.1 Motivation

Implementing a game on FPGA goes beyond entertainment — it is an excellent vehicle for mastering hardware design. The project requires students to think in terms of hardware signals and timing rather than sequential software instructions. Key motivations include:

- Gaining practical experience with Verilog HDL and the Vivado toolchain.
- Understanding VGA display interfacing, a fundamental skill in embedded display systems.
- Designing and verifying Finite State Machines (FSMs) for real-time logic.
- Learning clock domain management and synchronous design practices.
- Producing a tangible, interactive result that validates theoretical learning.

1.2 Scope of Project

The project covers the full design cycle from specification to hardware implementation. The scope includes:

- Design of a multi-module Verilog system for game logic and display.
- Implementation of a 640×480 @ 60 Hz VGA display interface.
- Real-time directional control using on-board push buttons.
- Pseudo-random food generation using a Linear Feedback Shift Register (LFSR).
- Collision detection with borders and snake body.

- Score tracking and display on the seven-segment display.
- Synthesis, implementation, and bitstream generation using Xilinx Vivado.

1.3 Organization of Report

- **Chapter 2:** Literature survey summarising related FPGA game implementations.
- **Chapter 3:** Details of all software and hardware tools used.
- **Chapter 4:** Full system design, module descriptions, and working principle.
- **Chapter 5:** Experimental results with observations.
- **Chapter 6:** Conclusion and future scope.

Chapter 2
Literature Survey

Several prior works on FPGA-based game implementations were reviewed to inform the design decisions of this project.

2.1 FPGA-Based Snake Game on VGA (Basys 3)

A Verilog implementation of the classic Snake game for the Basys 3 FPGA family was reviewed. In that implementation, the game is controlled by on-board arrow buttons and outputs to a VGA display, with the score shown on the seven-segment display.

- **Methodology:** Separate Verilog modules for VGA controller, game logic, and score display.
- **Result:** Successfully renders snake movement and food consumption on VGA at 640×480.
- **Limitation:** Does not include a timed game mode or pause functionality.

2.2 Snake Game on Basys 3 with Timed Mode

Another referenced project implemented Snake on the Basys 3 board with an added timed game mode. A switch enables a 60-second countdown during which the player must collect 10 targets. This project influenced our score display design.

- **Methodology:** FSM-based game control with a dedicated timer module using the 100 MHz clock.
- **Result:** Two game modes (free play and timed) operational on Basys 3 hardware.
- **Limitation:** No pseudo-random food placement; food appears at fixed positions.

2.3 FPGA Snake with VGA and SNES Controller

A more advanced implementation explored using a Super Nintendo Entertainment System (SNES) controller as input via GPIO pins. This work highlighted the flexibility of FPGAs for peripheral interfacing and influenced the modular structure adopted in our project.

- **Methodology:** Hardware-based PLL for clock generation; tile array for snake body tracking.

- **Result:** Fully functional game with hardware controller input.
- **Limitation:** Requires external DAC and non-standard controller hardware.

Chapter 3
Software and Hardware Used

3.1 Software

3.1.1 Xilinx Vivado Design Suite

Xilinx Vivado Design Suite is the primary EDA (Electronic Design Automation) tool used in this project. It supports the complete FPGA design flow for Artix-7 and other Xilinx devices, including HDL entry, simulation, synthesis, implementation, and bitstream generation.

3.1.2 ModelSim / Vivado Simulator

Testbench simulations were conducted to verify individual modules including the VGA controller, clock divider, and game FSM. Simulation confirmed that timing matched VGA standards before hardware testing.

3.2 Hardware

3.2.1 Basys 3 FPGA Board

The Basys 3 board from Digilent serves as the hardware platform for this project. Key peripherals used in this project include:

- 4 push buttons (BTNL, BTNR, BTNU, BTND) for directional control of the snake.
- VGA port (15-pin D-sub connector) with 12-bit colour.
- Four-digit seven-segment display for real-time score output.
- Slide switches for game reset and mode selection.
- USB-JTAG interface for programming via Vivado.

3.2.2 VGA Monitor

A standard VGA-compatible monitor is used to display the game output. The display resolution is 640×480 pixels at a refresh rate of 60 Hz.

Table 3.1: Hardware and Software Specifications

Component	**Specification / Detail**
FPGA	Xilinx Artix-7 XC7A35T-1CPG236C
Board	Digilent Basys 3
Clock	100 MHz on-board oscillator
VGA Output	640×480 @ 60 Hz, 12-bit colour
Input	4 push buttons (directional), slide switches
Score Display	4-digit seven-segment display
HDL Language	Verilog / SystemVerilog
EDA Tool	Xilinx Vivado 2022.x

Chapter 4

System Design and Working Principle

4.1 System Architecture

The system is designed using a modular Verilog approach. The major modules are:

- `top.v` — Top-level module; connects all sub-modules.
- `clk_div.v` — Clock divider; derives 25 MHz VGA clock from 100 MHz system clock.
- `vga_controller.v` — Generates HSYNC, VSYNC signals and pixel coordinates (x, y).
- `game_logic.v` — Core FSM; manages snake position, direction, and game state.
- `collision.v` — Detects collisions with walls and the snake's own body.
- `food_gen.v` — Generates pseudo-random (x, y) coordinates for food placement.
- `renderer.v` — Determines RGB pixel colour based on game objects.
- `score_display.v` — Drives the seven-segment display with the current score.

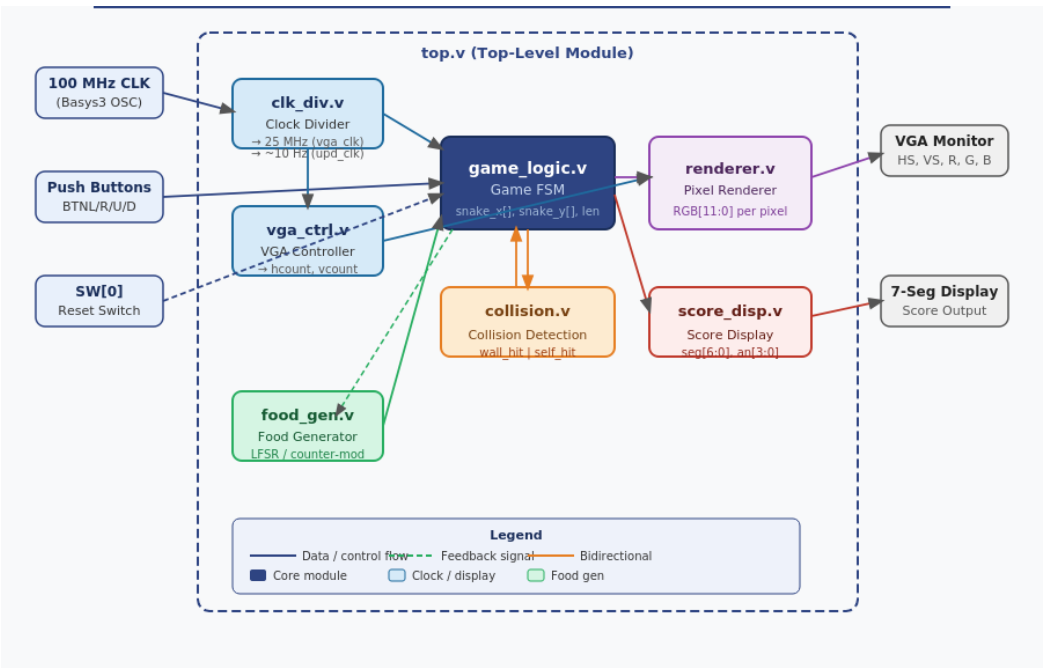

Figure 4.1: System Architecture: Module Interconnection Diagram

4.2 Module Descriptions

4.2.1 Clock Divider

The Basys 3 board provides a 100 MHz system clock. The VGA standard requires a 25 MHz pixel clock. A separate, much slower update clock (approximately 5–25 Hz) is derived to govern the snake's movement speed.

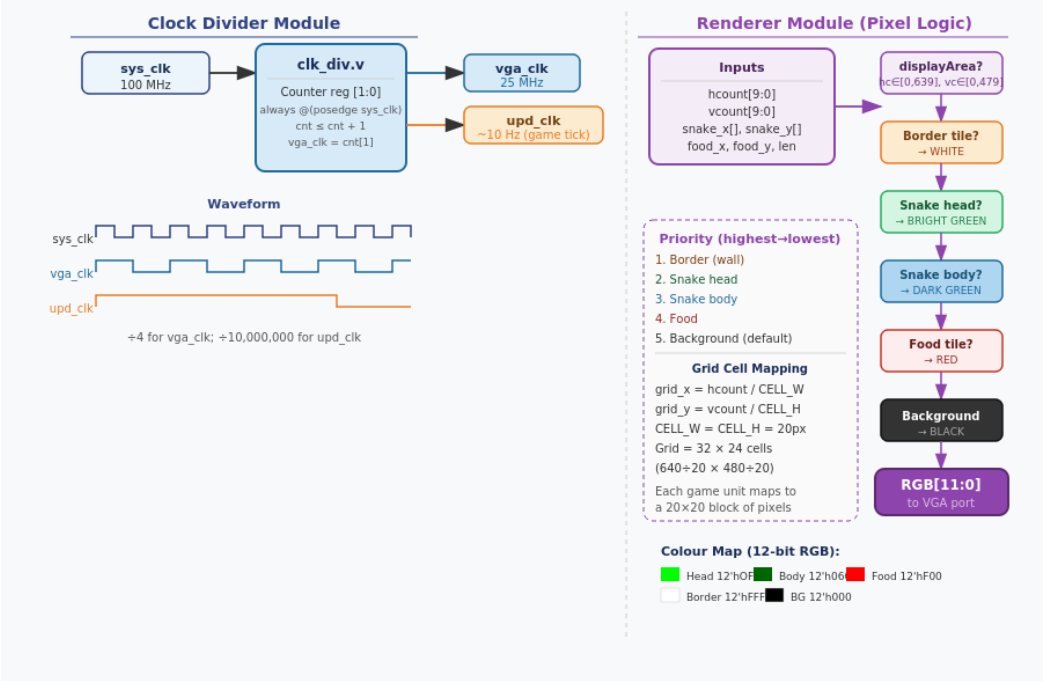

Figure 4.2: VGA Renderer Logic & Clock Divider Design

4.2.2 VGA Controller

The VGA controller generates the horizontal synchronisation (HSYNC) and vertical synchronisation (VSYNC) signals and outputs the current pixel column (x) and row (y) counters.

Table 4.1: VGA Timing Parameters

Parameter	Horizontal	Vertical
Active Display	640 pixels	480 lines
Front Porch	16 pixels	10 lines
Sync Pulse	96 pixels	2 lines
Back Porch	48 pixels	33 lines

4.2.3 Game Logic FSM

The game logic is implemented as a Finite State Machine with primary states: IDLE, RUNNING, FOOD_EATEN, COLLISION, and GAME_OVER. The snake body is tracked using a shift-register-style array of coordinate pairs.

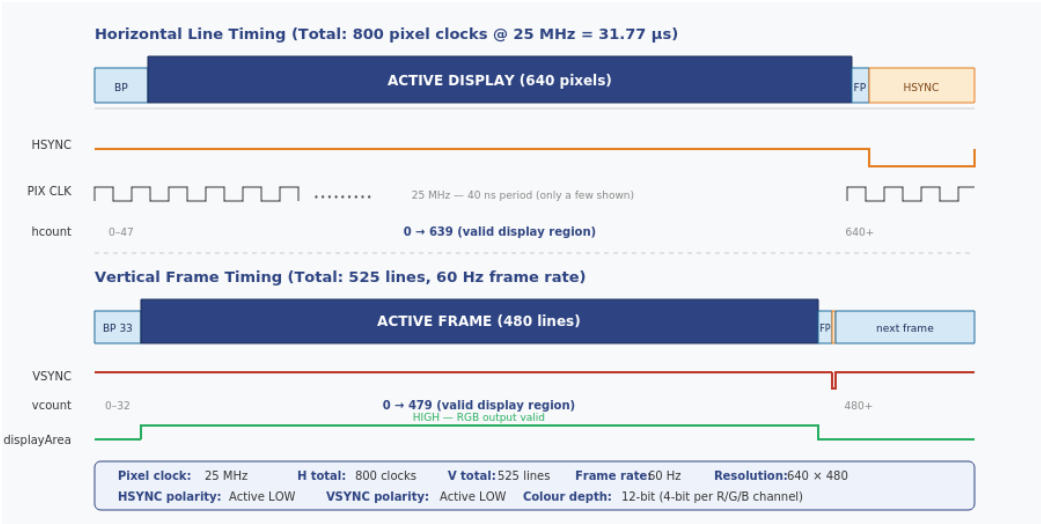

Figure 4.3: VGA 640 × 480 @ 60 Hz Timing Diagram

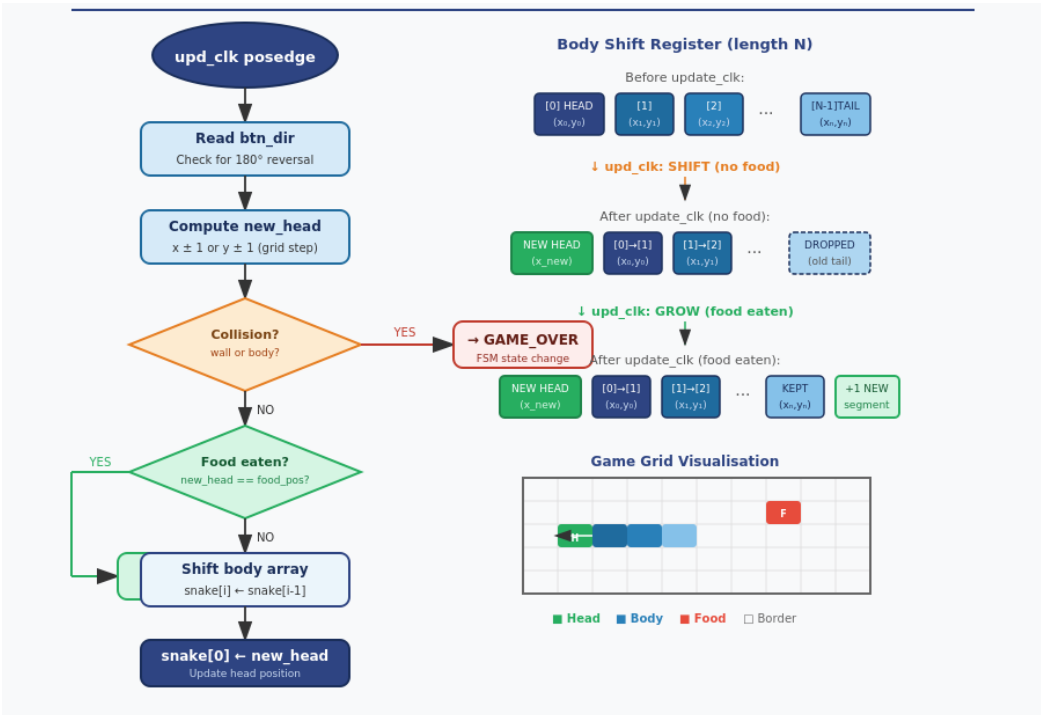

Figure 4.4: Snake Movement & Body Update Flowchart

4.2.4 Collision Detection

Collision detection checks two conditions: wall collision and self-collision (head matching any body segment).

4.2.5 Food Generator

Food is placed at pseudo-random positions using a Linear Feedback Shift Register (LFSR) whenever the current food is consumed.

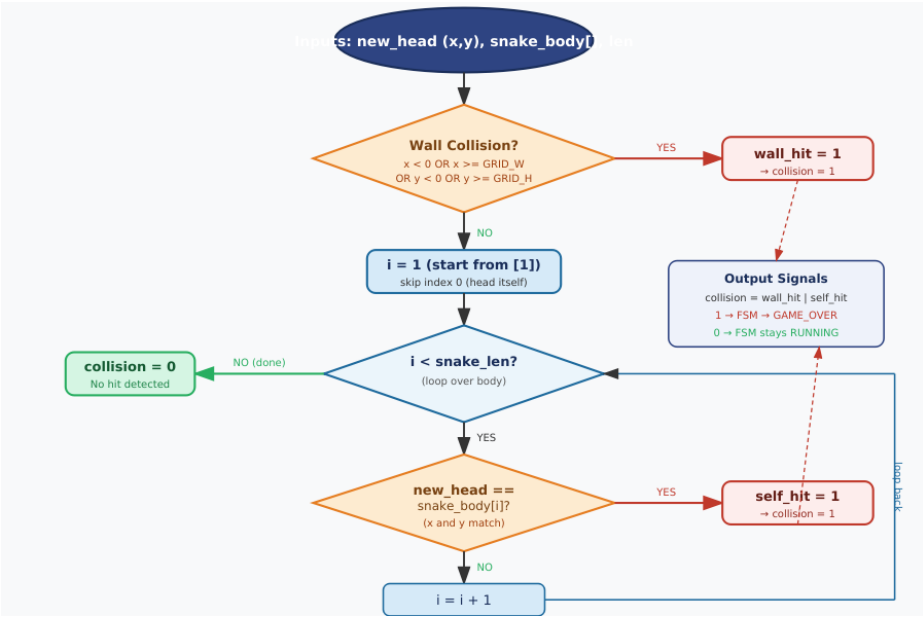

Figure 4.5: Collision Detection Module Flowchart

4.3 VGA Display Interface

The renderer module computes RGB values. Snake head (bright green), body segments (darker green), food (red), border walls (white), and background (black) are rendered based on (x, y).

4.4 Game Logic and State Machine

Direction input is captured from 4 buttons. Debouncing ensures accuracy. The FSM prevents 180-degree reversals.

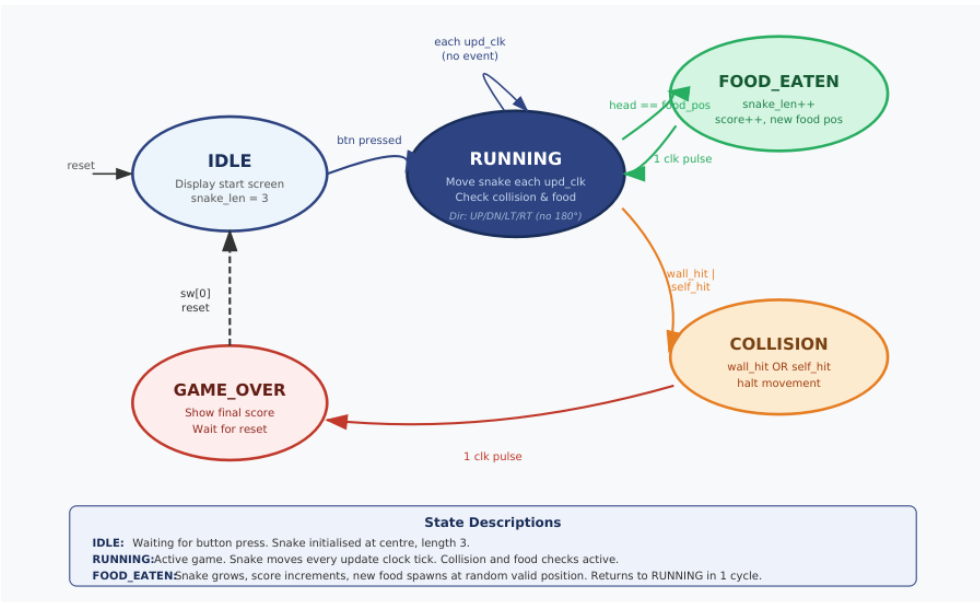

Figure 4.6: Game Logic Finite State Machine (FSM)

4.5 Verilog Implementation on Basys 3

The top-level I/O interface maps system signals to physical pins on the Basys 3 board. The top-level I/O interface maps system signals to physical pins on the Basys 3 board. Each signal is described below:

Table 4.2: Top-Level I/O Interface

Signal	Direction	Description
clk	Input	100 MHz system clock from oscillator
btnL, btnR, btnU, btnD	Input	Directional push buttons for control
sw[0]	Input	Reset switch
vga_hs, vga_vs	Output	VGA sync signals
vga_r, vga_g, vga_b	Output	4-bit RGB colour channels
seg[6:0], an[3:0]	Output	Seven-segment display signals for score

- **clk (Input):** 100 MHz onboard crystal oscillator clock driving all synchronous logic. A clock divider divides this by 4 to generate the 25 MHz pixel clock required by the VGA controller.
- **btnL, btnR, btnU, btnD (Input):** Four directional push buttons controlling snake movement. Each passes through a debouncer module to eliminate mechanical bounce and ensure one clean pulse per press.
- **sw[0] (Input):** Active-high reset switch. When asserted, all game state is cleared — the snake returns to its starting position, the score resets to zero, and food repositions to its default location.
- **vga_hs, vga_vs (Output):** Horizontal and vertical sync signals following the 640x480 at 60 Hz VGA standard, generated by the VGA controller using pixel and line counters driven by the 25 MHz pixel clock.
- **vga_r, vga_g, vga_b (Output):** 4-bit RGB colour channels giving 16 intensity levels per channel and 4096 total colours. The VGA controller assigns colours per pixel — bright green for the snake head, dark green for the body, red for food, white for the border, and black for the background.
- **seg[6:0], an[3:0] (Output):** Seven-segment display signals for showing the score. `seg[6:0]` controls the active segments while `an[3:0]` selects the active digit. The seg7_controller multiplexes all four digits rapidly, making them appear simultaneously lit.

Chapter 5
Results and Analysis

5.1 Waveform Verification

Sync signals were verified to match the 60 Hz timing requirements.

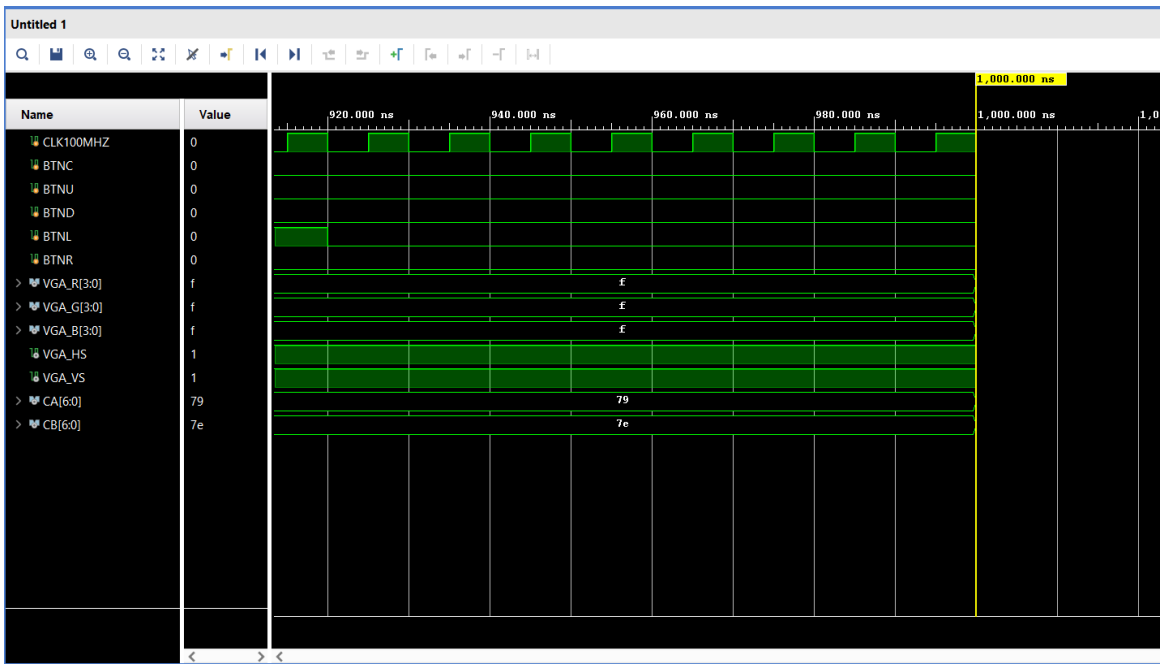

Figure 5.1: Behavioral Simulation Waveforms showing VGA Sync Timing

5.2 Hardware Implementation Results

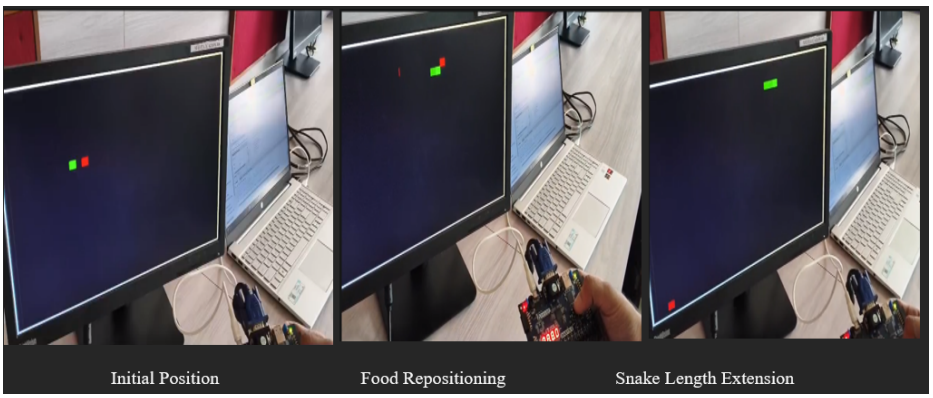

Figure 5.2: Snake Game Gameplay Output on VGA Monitor

Hardware testing confirmed that the modular Verilog design translates correctly to physical gates.

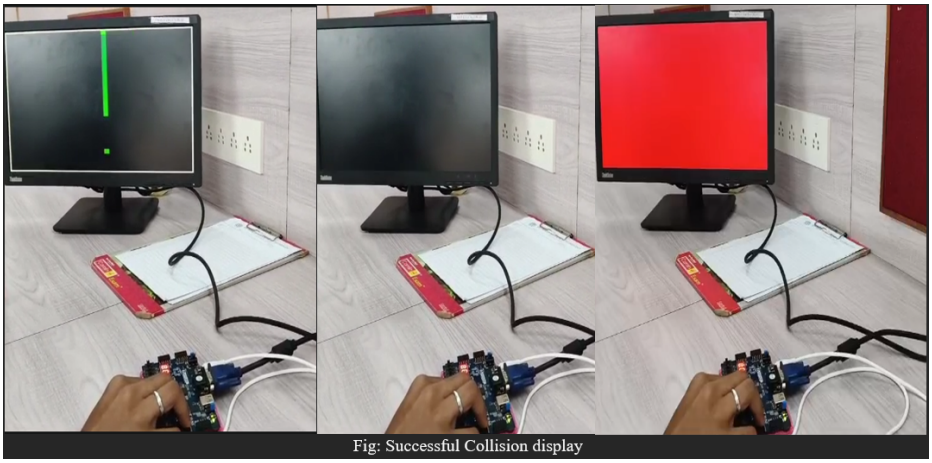

Figure 5.3: Final Hardware Setup showing Basys 3 Board and Monitor

5.3 Resource Utilisation

Following the implementation phase, the Vivado Power and Utilization reports provided the following data regarding FPGA resource consumption:

Table 5.1: Final FPGA Resource Utilisation

Resource	**Used**	**Utilisation (%)**
Slice LUTs	382	∼1.8%
Slice Registers	424	∼1.0%
Bonded IOBs	18	∼8.6%
On-Chip Power	0.072 W	—

Note: Power and Utilization values are derived from the Xilinx Vivado Implemented Design report.

Chapter 6
Conclusion and Future Scope

6.1 Conclusion

This project successfully implemented the Snake game on a Basys 3 FPGA board using

Verilog HDL, with output on a VGA display and score tracking on the seven-segment

display. All key game mechanics — snake movement, food consumption, growth, collision

detection, and game-over logic — were implemented entirely in hardware using

synchronous digital design principles.

The project provided deep practical experience in several important areas of digital

design: VGA timing and display interfacing, FSM design and verification, clock domain

management, modular Verilog coding, and the use of the Xilinx Vivado toolchain for

synthesis and implementation. The design was verified through both simulation and

hardware testing, with results matching expected behaviour.

A key takeaway is that FPGA-based design, while more complex to develop than a

microcontroller-based software solution, offers deterministic real-time performance and

true hardware parallelism. This makes it highly suitable for time-critical interactive

applications such as games, and scales effectively to larger and more complex designs.

6.2 Future Scope

The current implementation can be extended in several directions:

- **Difficulty levels:** Implement variable game speed using DIP switches to select between multiple update clock frequencies, providing adjustable difficulty.
- **Colour enhancements:** Use the full 12-bit VGA colour depth to add gradient effects, animated food, or background patterns.
- **Keyboard input:** Interface a PS/2 keyboard to the Basys 3 GPIO pins for standard arrow-key control, improving user experience.
- **Multiple food items:** Track multiple simultaneous food positions to increase complexity and scoring opportunities.

- **High-score memory:** Utilise the FPGA's block RAM or external SPI flash to persist the high score across game sessions.
- **Sound effects:** Generate simple tones through a GPIO-connected piezo buzzer to add audio feedback for food collection and game-over events.
- **Larger grid:** Scale the design to a larger display resolution (800×600 or 1024×768) with a finer grid for a more detailed game field.
- **Multiplayer mode:** Implement two-player Snake by adding a second snake controlled by different button inputs, with inter-snake collision detection.

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